

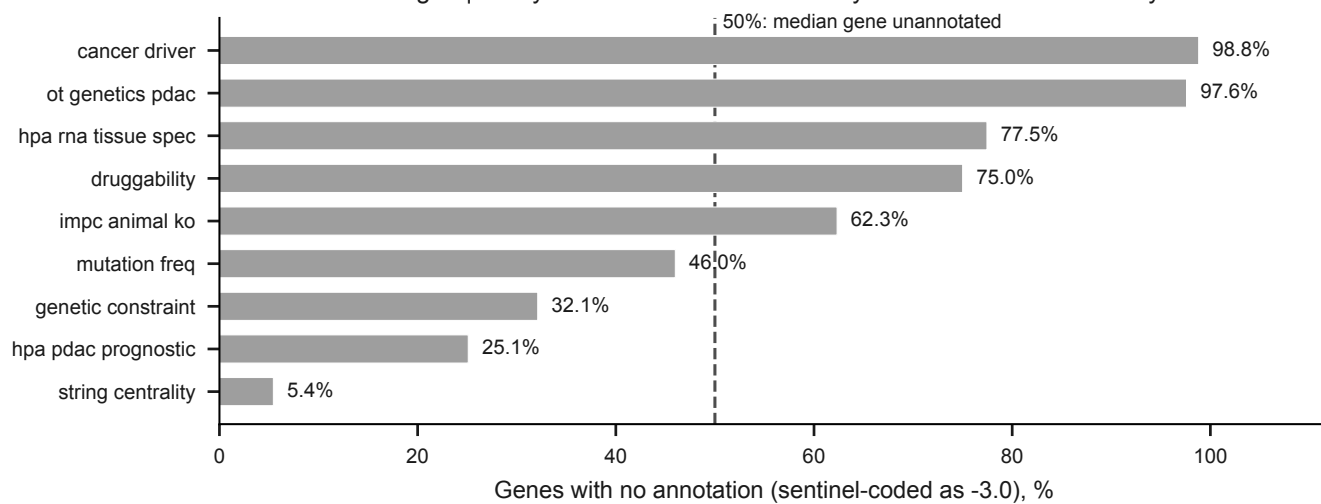
Supplementary Information

Auditing when evidence integration improves therapeutic target prioritization

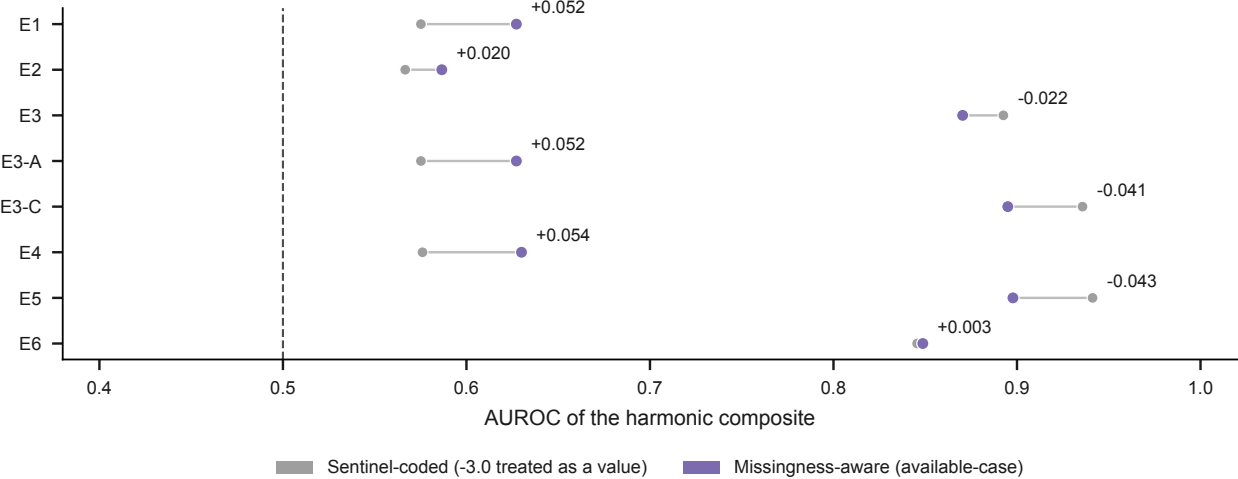
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Extended Data Fig. 1 | Study-bias and annotation-density audit of network centrality

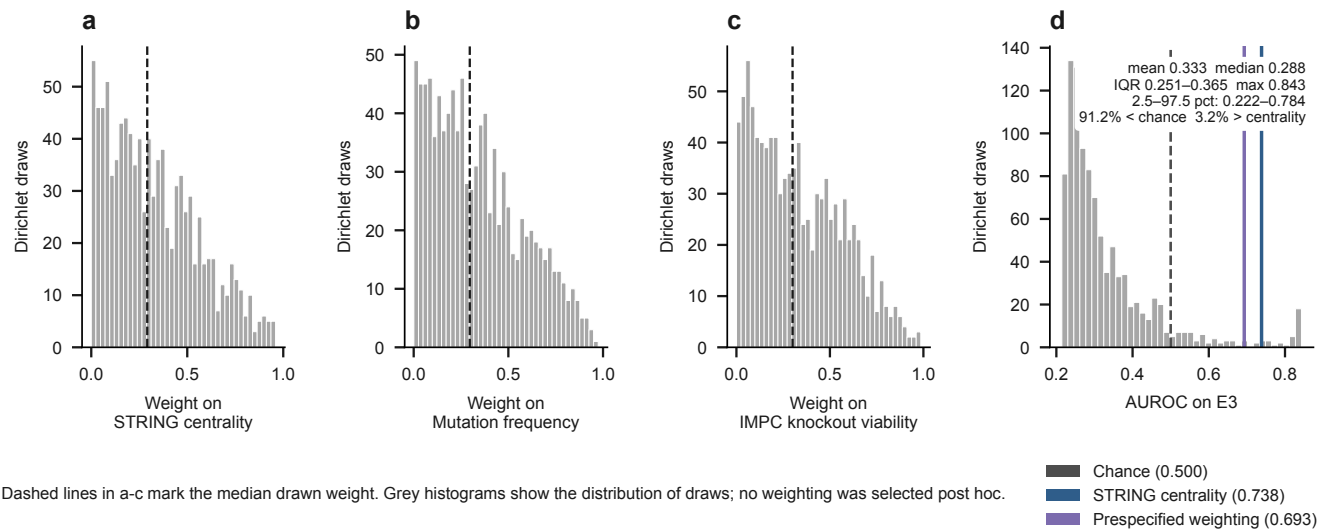


Extended Data Fig. 2 | Missingness-encoding sensitivity, comparing available-case handling with missingness-aware analysis

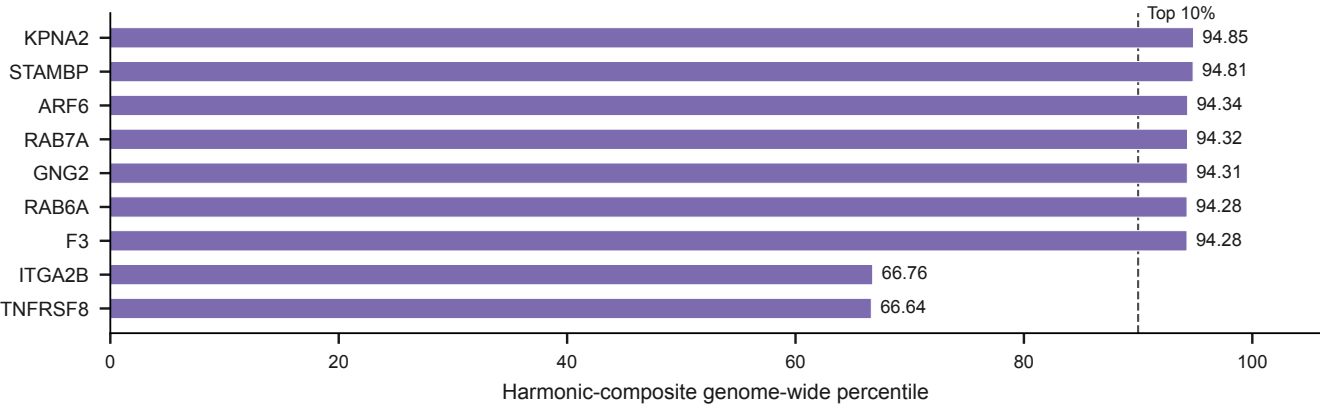


Labels give the signed AUROC change (missingness-aware minus sentinel-coded). Dashed line marks chance (0.5).

Extended Data Fig. 3 | Full Dirichlet weight-space distribution



Extended Data Fig. 4 | Nine candidate genes as prospective computational hypotheses



These are prospective computational hypotheses, not validated therapeutic targets; no experimental confirmation is claimed.

Supplementary Table 1 | Audit ledger documenting each P0/P1 item, severity, finding and action taken.

Item	Requirement	Status	Action taken / evidence
P0-1	E5/E6 source-overlap audit	Qualitative overlap established; exact gene-level overlap requires the released inputs.	`code/audit_overlap.py`; E6 definition-overlap shown in Fig. 6; qualitative overlap reported in Results, exact gene-level overlap computed by the audit script once the released inputs are present.
P0-2	Leakage-safe nested cross-validation	Protocol specified; not used in the primary benchmark	`code/audit_nested_cv.py`; protocol described in Methods & Supp. Table 5; the released benchmark does not perform endpoint-specific hyperparameter search.
P0-3	Centrality annotation-density / study-bias	Completed	`code/audit_centrality_bias.py`; ED Fig. 1; caveat in Discussion.
P0-4	E4 cross-context wording	Fixed in text	'demonstrates cross-context transfer but not fully independent external validity'.
P0-5	Three-class endpoint distinction	Fixed in text	'six primary evaluation endpoints' + 'two non-independent constructs'.
P0-6	Dirichlet full distribution	Real numbers reported	1,000 draws: mean 0.333, median 0.288, 91.2% below chance, 3.2% exceed centrality (0.738).
P0-7	Final clean rerun / provenance chain	Clean rerun from the released inputs reproduced all 104 scorer-endpoint benchmark cells to six decimals.	
P1-1	'tractability/druggability layer' canonical terminology	Fixed	Throughout Methods/Results.
P1-4	E6 'pharmacological-response proxy' wording	Fixed	Results 'Cross-context transfer-' section; Fig. 6a label + Supplementary Table 2 entry.
P1-5	'cannot be attributed to integration alone' (RF gain)	Fixed	Results, single-layer-vs-integration paragraph.
P1-7	'Primary contrasts were pre-specified'	Fixed	Methods · Statistics.
P1-9	Fig. 6 redesign; candidates · ED Fig. 4	Fixed	Fig. 6 = pharmacological stress test (2 panels); candidates in ED Fig. 4.

Supplementary Table 2 | Endpoint provenance and independence matrix.

ID	Endpoint	Positive definition	Positives	Provenance class	Label external to evidence base (external-label / constructed / circular)
E1	PDAC-wide dependency	DepMap 23Q2 Chronos gene effect in pancreatic lines below the essentiality threshold	4584	external-label	yes
E2	PDAC-enriched dependency	upper quartile of the PDAC-vs-other selectivity statistic among E1 positives	1147	non-independent (stratification of E1)	yes (nested in E1)
E3	essential-and-druggable construct	E1 positive AND annotated druggable (conjunctive benchmark construct)	1159	constructed/circular	no
E3-A	essentiality-only control	E1 positive with the druggability conjunct removed	4584	non-independent (identical to E1)	yes (identical to E1)
E3-C	circular druggability control	chemical-tractability layer used directly as the label	5188	constructed/circular	no
E4	CRC zero-shot transfer	DepMap 23Q2 Chronos gene effect in colorectal lines, scoring not refitted	4608	external-label	yes
E5	historical clinical-target concordance	nominal primary target of an agent entering pancreatic-cancer clinical development	35	external-label	yes (overlaps tractability)
E6	PDAC pharmacological-response proxy	nominal target of a GDSC compound in the most sensitive tertile across pancreatic lines	32	external-label	yes (overlaps tractability)

The six primary evaluation endpoints = E1, E4, E5, E6 (external-label) + E3, E3-C (constructed diagnostics). E2 and E3-A are reported for continuity and excluded from independent-endpoint counts.

Supplementary Table 3 | Evidence-layer overlap with E5 and E6 labels.

Endpoint	Label source	Overlap with tractability/druggability layer	Evidence
E5	nominal target of agents entering pancreatic clinical development (ClinicalTrials.gov, frozen 2024-12-31)	Yes: clinical development history favours druggable targets	qualitative overlap reported in manuscript; Gene-level overlap is not reported because the corresponding third-party target lists are not redistributed; the qualitative overlap relevant to endpoint provenance is reported here and in the manuscript.
E6	nominal target of GDSC compounds in the most sensitive tertile (125 compounds, 29 lines)	Yes: GDSC compounds are, by construction, druggable	qualitative overlap reported in manuscript; Gene-level overlap is not reported because the corresponding third-party target lists are not redistributed; the qualitative overlap relevant to endpoint provenance is reported here and in the manuscript.

Honesty note: the gene-level overlap counts require the raw E5/E6 target lists and the tractability layer, which are not redistributed in this repository. The qualitative overlap is reported in the manuscript (Results, first section); the quantitative overlap is computed by code/audit_overlap.py once the inputs are released.

Supplementary Table 4 | AUROC (AUPRC) for each scorer across endpoints.

Scorer	E1	E2	E3	E3-A	E3-C	E4	E5	E6
Mutation frequency	0.499 (0.217)	0.491 (0.055)	0.520 (0.058)	0.499 (0.217)	0.563 (0.282)	0.497 (0.218)	0.702 (0.115)	0.567 (0.002)
Genetic constraint	0.524 (0.218)	0.496 (0.051)	0.559 (0.058)	0.524 (0.218)	0.643 (0.310)	0.530 (0.224)	0.389 (0.001)	0.371 (0.001)
Cancer-driver annotation	0.507 (0.225)	0.506 (0.057)	0.521 (0.066)	0.507 (0.225)	0.511 (0.260)	0.507 (0.227)	0.737 (0.035)	0.635 (0.011)
Druggability	0.507 (0.231)	0.526 (0.060)	0.915 (0.260)	0.507 (0.231)	1.000 (1.000)	0.506 (0.232)	0.808 (0.008)	0.780 (0.007)
STRING centrality	0.695 (0.399)	0.601 (0.081)	0.738 (0.175)	0.695 (0.399)	0.620 (0.356)	0.702 (0.408)	0.986 (0.135)	0.910 (0.027)
Arithmetic mean	0.580 (0.253)	0.561 (0.066)	0.832 (0.197)	0.580 (0.253)	0.874 (0.760)	0.580 (0.254)	0.894 (0.137)	0.732 (0.051)
Rank aggregation	0.533 (0.241)	0.518 (0.061)	0.636 (0.094)	0.533 (0.241)	0.679 (0.424)	0.534 (0.242)	0.854 (0.018)	0.786 (0.013)
Weighted rank aggregation	0.634 (0.311)	0.575 (0.073)	0.835 (0.260)	0.634 (0.311)	0.808 (0.641)	0.638 (0.316)	0.956 (0.234)	0.906 (0.105)
ECS (multiplicative)	0.415 (0.179)	0.469 (0.049)	0.693 (0.080)	0.415 (0.179)	0.717 (0.346)	0.415 (0.180)	0.825 (0.009)	0.705 (0.004)
Harmonic mean	0.575 (0.254)	0.567 (0.068)	0.893 (0.245)	0.575 (0.254)	0.936 (0.873)	0.576 (0.255)	0.941 (0.207)	0.846 (0.078)
Logistic regression	0.679 (0.338)	0.603 (0.077)	0.931 (0.371)	0.679 (0.338)	1.000 (1.000)	0.683 (0.344)	0.982 (0.291)	0.910 (0.021)
Elastic net	0.671 (0.327)	0.594 (0.074)	0.927 (0.333)	0.671 (0.327)	1.000 (1.000)	0.675 (0.332)	0.899 (0.168)	0.481 (0.001)
Random forest	0.750 (0.490)	0.631 (0.089)	0.942 (0.486)	0.750 (0.490)	1.000 (1.000)	0.756 (0.499)	0.942 (0.250)	0.913 (0.017)

Each cell reports AUROC (AUPRC).

Supplementary Table 4 (continued) | Pairwise AUROC differences with 95% CIs and DeLong P values for selected comparisons.

Endpoint	Comparison	AUROC difference (95% CI)	DeLong P
E1	Harmonic mean	-0.120 (-0.129 to -0.110)	1.81e-141
E1	Random forest	0.055 (0.049 to 0.062)	1.23e-66
E1	ECS (multiplicative)	-0.280 (-0.294 to -0.265)	< 1e-300
E2	Harmonic mean	-0.035 (-0.052 to -0.018)	5.38e-05
E2	Random forest	0.030 (0.013 to 0.046)	0.00033
E2	ECS (multiplicative)	-0.133 (-0.159 to -0.106)	4.4e-23
E3	Harmonic mean	0.154 (0.141 to 0.168)	1.68e-115
E3	Random forest	0.204 (0.190 to 0.218)	1.08e-183
E3	ECS (multiplicative)	-0.046 (-0.062 to -0.029)	5.45e-08
E3-A	Harmonic mean	-0.120 (-0.129 to -0.110)	1.81e-141
E3-A	Random forest	0.055 (0.049 to 0.062)	1.23e-66
E3-A	ECS (multiplicative)	-0.280 (-0.294 to -0.265)	< 1e-300
E3-C	Harmonic mean	0.316 (0.308 to 0.324)	< 1e-300
E3-C	Random forest	0.380 (0.371 to 0.389)	< 1e-300
E3-C	ECS (multiplicative)	0.097 (0.085 to 0.109)	2.88e-59
E4	Harmonic mean	-0.126 (-0.135 to -0.116)	8.04e-157
E4	Random forest	0.054 (0.048 to 0.061)	2.35e-65
E4	ECS (multiplicative)	-0.287 (-0.300 to -0.273)	< 1e-300
E5	Harmonic mean	-0.045 (-0.085 to -0.013)	0.0167
E5	Random forest	-0.044 (-0.098 to 0.001)	0.0813
E5	ECS (multiplicative)	-0.163 (-0.245 to -0.093)	3.71e-05
E6	Harmonic mean	-0.066 (-0.134 to -0.004)	0.0533
E6	Random forest	0.003 (-0.032 to 0.040)	0.864
E6	ECS (multiplicative)	-0.208 (-0.330 to -0.087)	0.000883

Supervised learners (logistic regression, elastic net, random forest) use out-of-fold predictions from five-fold stratified cross-validation. Learners on E3/E3-C are circular by construction and reported for completeness. 95% percentile CIs from 2,000 stratified bootstrap resamples are in source_data.csv.

Supplementary Table 5 | Nested cross-validation hyperparameters and protocol.

Component	Specification
Outer loop	five-fold stratified CV (seed fixed)
Inner loop	three-fold grid search over hyperparameters (fixed candidate grid)
Candidates (elastic net)	$\text{l1_ratio} \cdot \{0.1, 0.5, 0.9\}$; $C \cdot \{0.01, 0.1, 1.0\}$
Candidates (random forest)	$n_estimators \cdot \{100, 500\}$; $\text{max_depth} \cdot \{\text{None}, 8, 16\}$
Candidates (logistic regression)	$C \cdot \{0.01, 0.1, 1.0, 10.0\}$
Evaluation	out-of-fold predictions only; no endpoint used for tuning
Circularity guard	learners on E3/E3-C are circular by construction; reported for completeness only
Runtime status	Protocol specified in code/audit_nested_cv.py; not used in the primary benchmark